Tahtabir Rahman

+919546079819 | India | tahtabirr@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

Data Engineer with **4+ years** of experience in data engineering and analytics, building reliable **ETL/ELT pipelines** that turn raw data from multiple sources into governed, analytics-ready datasets. Currently building a healthcare **data lake platform on AWS** using **Python, PySpark, SQL, AWS Glue, S3, Athena, Lambda, and Step Functions**, with layered raw-staging-curated design, metadata-driven transformations, and orchestrated, monitored workflows. Strong in data quality, reconciliation, and pipeline reliability, and proven at automating manual data work (**30%** reduction in manual data handling). Partners with engineering and business teams to deliver dependable data that supports reporting, BI, and ML.

Technical Skills

Languages: Python (Pandas, NumPy, PySpark), SQL
AWS Cloud: S3, Glue, Athena, Lambda, Step Functions, CloudWatch, IAM, Glue Data Catalog
Data Pipelines & ETL/ELT: PySpark ETL, Apache Spark, Batch Processing, Data Lake (Raw/Staging/Curated), Metadata-Driven Pipelines, Pipeline Automation
Data Quality: Validation, Cleansing, Reconciliation, Deduplication, Schema Enforcement, Exception Handling, Profiling
Data Modeling & SQL: Data Modeling, Joins, CTEs, Window Functions, Partitioning, Bucketing, Query Optimization
Databases & Formats: MySQL, DynamoDB, Parquet, CSV, JSON
DevOps & Version Control: Git, GitHub, CI/CD
BI & Reporting: Power BI, Tableau, Advanced Excel, KPI Dashboards
Other Cloud: Microsoft Azure (Data Factory, Databricks) – foundational

Experience

Data Engineer India
S-Cube Solutions Private Limited *Nov 2025 – Present*
Project: Healthcare Data Lake Platform
Tech Stack: Python, PySpark, MySQL, SQL, DynamoDB, AWS (S3, Glue, Athena, Lambda, Step Functions, CloudWatch, IAM), Git

- **Overview:** Built a cloud-native batch data platform on AWS that consolidates raw healthcare data from multiple client sources into governed, analytics-ready datasets for BI and ML use, on a layered S3 data lake (raw → staging → curated).
- **Delivered analytics-ready healthcare data:** Built PySpark ETL pipelines on AWS Glue that cleanse, conform, and load multi-source data into partitioned Parquet datasets in S3, queryable through Athena.
- **Unified heterogeneous sources:** Ingested data from MySQL and DynamoDB sources into the S3 raw layer using Glue connectors, handling incremental loads and schema changes.
- **Accelerated source onboarding:** Enabled faster onboarding of new data sources with a metadata-driven transformation layer that applies source-to-target mappings and business rules with minimal code changes.
- **Protected data trust:** Embedded quality controls at every stage: schema enforcement, deduplication, null and type validation, and record-count reconciliation between layers.
- **Strengthened reliability:** Orchestrated workflows with Step Functions and Lambda, added retry and failure handling, and set up CloudWatch alarms for monitoring and alerting.
- **Improved performance and cost:** Reduced Athena query runtime and scan costs through partitioning, bucketing, predicate pushdown, and window and analytical functions, and tuned Glue job configuration (worker types, DPU allocation, join strategies) to improve throughput and compute spend.
- **Enabled business reporting:** Maintained Glue Data Catalog and Athena schemas and views as a governed serving layer for reporting teams.
- **Ensured controlled releases:** Shipped code through Git-based version control and CI/CD for repeatable, low-risk releases.

Analyst Noida, India

HCL Tech *Apr 2022 – Oct 2025*

Project: IT Operations Data Pipelines & Reporting

Tech Stack: Python, SQL, Data Modeling, AWS S3, ServiceNow

- **Automated recurring data pipelines:** Built Python and SQL pipelines that replaced manual data preparation, reducing manual data handling by 30% and improving processing consistency.
- **Built ETL workflows:** Developed Python scripts and SQL transformations to extract data from ServiceNow and operational sources, then cleanse, transform, validate, and consolidate it into analytics-ready datasets for networking, cybersecurity, and IT operations teams.
- **Designed SQL data models:** Structured operational and service-management data into consistent, reusable tables and views to support recurring reporting and compliance requirements.
- **Protected data integrity:** Implemented validation and reconciliation checks to catch missing, duplicate, and inconsistent records across operational and compliance datasets, and resolved discrepancies through root-cause analysis.
- **Improved performance and storage:** Optimized SQL queries and Python scripts for extraction and transformation, and organized data in AWS S3 for structured storage and downstream consumption, delivering datasets within SLAs.

Projects

Vendor Data Integration & ETL Pipeline

Python, SQL, Pandas, SQLAlchemy, ETL, Power BI, Git/GitHub

- **Built an ETL pipeline for 2.3M+ records:** Engineered a Python, Pandas, and SQLAlchemy pipeline that ingests 6 source datasets into a relational database, using chunked loading to handle the 2.3M-row sales file reliably.
- **Unified fragmented data:** Used SQL multi-table joins to combine purchase, sales, freight, vendor, and brand data into one analytics-ready dataset of **10,692** vendor-brand records across 18 fields, with zero null values.
- **Found and fixed a data integrity defect:** Traced invalid profit margin values in 178 zero-sales records back to the transformation logic and corrected the rules, so financial metrics stayed accurate and consistent downstream.
- **Validated and preserved data:** Applied cleansing, reconciliation, and validation checks, and flagged 698 dead-stock outliers (6.5% of records), filtering them out of margin averages instead of deleting them.
- **Delivered business-ready output:** Built reusable Python and SQL logic for vendor performance and inventory metrics, feeding a Power BI dashboard that tracks **$451.6M** in sales, a **28.7%** profit margin, and **$939.7K** in unsold capital.

E-Commerce Data Integration Pipeline

Python, SQL, PySpark, Azure Data Factory, Azure Databricks, Git

- **Unified e-commerce data:** Built ETL pipelines that consolidate product catalog, customer order, and sales transaction data into structured, analytics-ready datasets.
- **Ensured data consistency:** Applied cleansing, validation, and reconciliation (Python, SQL, PySpark) so related datasets agree across orders and sales.
- **Automated on Azure:** Orchestrated ingestion and transformation with Azure Data Factory and Databricks, with all code version-controlled in Git.

Returns & Refunds Data Pipeline

Python, SQL, PySpark, Azure Data Factory, Azure Databricks, Git

- **Consolidated returns data:** Integrated product returns, refund transactions, and return-status records into consistent operational datasets.
- **Improved tracking reliability:** Automated cleansing, transformation, and validation to flag inconsistent records before they affect returns reporting.
- **Enabled operational reporting:** Built repeatable workflows that deliver validated datasets for downstream reporting.

Payment Transaction Data Processing

Python, SQL

- **Consolidated payment data:** Developed Python and SQL logic to combine payment transactions, customer records, and refund data into structured datasets for financial reporting.
- **Protected financial accuracy:** Implemented transaction validation and reconciliation checks to catch inconsistencies and improve data reliability.
- **Prepared analysis-ready data:** Refined SQL transformations to produce consistent financial datasets for downstream analysis.

Education

Maulana Azad College of Engineering & Technology, Patna 2017 – 2021
Bachelor of Technology in Computer Science and Engineering

Certifications

AWS Cloud Practitioner Bootcamp (HCL Tech) Link
Power BI (PL-300) (HCL Tech) Link
Foundations: Data, Data, Everywhere (Coursera) Link
Ask Questions to Make Data-Driven Decisions (Coursera) Link
Machine Learning (Coursera) Link
Microsoft Excel (Coursera) Link

Extra Contributions

Volunteering: Received multiple appreciations from organizations for exceeding project expectations (2022–2025)